

ADMIN DASHBOARD REQUIREMENTS DOCUMENT

V2 Application — No Withdrawal / No Transfer Functionality

1. Document Purpose

This document defines the functional requirements for the Admin Dashboard of the V2 financial/VTU application. The scope is limited to the features currently supported by the application. **Withdrawals and user-to-user transfers are intentionally excluded.**

2. Admin Access & Role-Based Permissions

- Secure admin login separate from normal user authentication.
- Role-based access control (RBAC) for different administrator permissions.
- Permissions for viewing users, managing balances, viewing transactions, managing VTU services, reviewing risk alerts, and managing settings.
- Secure admin sessions and authentication.
- Record sensitive administrator actions in audit logs.
- Prevent normal users from accessing admin APIs or dashboard pages.

3. Dashboard / Overview

- Total registered users.
- Active users.
- Inactive users.
- Blocked users.
- New users registered today/this week/month.
- Total deposits.
- Total successful transaction value.
- Total pending transaction value.
- Total failed transaction value.
- Total transaction count.
- Successful, pending, and failed transaction counts.
- VTU provider wallet balance.
- Today's VTU spending.
- This month's VTU spending.
- Recent transactions.
- Recent deposits.
- Recent user registrations.
- System and risk alerts.
- Low VTU provider balance warning.

4. User Management

- View all registered users.
- Search by name, email, phone number, user ID, account number, or supported identifier.
- Filter by active, inactive, blocked, verified, unverified, and registration date.
- Open an individual user profile.
- View user's current balance.
- View user's complete ledger and transaction history.
- View deposits and VTU purchases.
- Block a user.
- Unblock/reactivate a user.
- Restrict account activity where supported.
- View relevant security/activity information.
- Add customer-support notes to a user account.

5. User Wallet & Ledger Management

- Add/fund money to a user's balance.
- Deduct/reduce money from a user's balance when authorized.
- Every manual balance adjustment must create a ledger entry.
- Require a reason/reference for each manual adjustment.
- Record the administrator who performed the adjustment.
- Record date and time.
- Record amount, adjustment type, and unique reference.
- Record balance before and after the adjustment.
- Use idempotency/reference checks to prevent duplicate adjustments.
- Do not delete financial ledger history; corrections should use reversal/adjustment entries.

6. Deposit Management

- View all deposits.
- Search deposits by transaction/reference ID, user, account number, amount, or date.
- Filter deposits by successful, pending, failed, reversed, or other supported status.
- Open complete deposit details.
- View payment-provider references where applicable.
- Investigate pending or failed deposits.
- Ensure confirmed deposits create the appropriate ledger entry.
- Prevent duplicate crediting of the same deposit through idempotency/reference validation.

7. Transaction Management

- View all transactions performed in the application.
- Search by transaction ID/reference, user ID, account number, phone number, or other identifiers.
- Filter by transaction type such as deposit, VTU purchase, refund, reversal, or admin adjustment.
- Filter by successful, pending, failed, cancelled, or reversed status.

- Filter by date range.
- Filter by amount range.
- Open complete transaction details.
- View timestamps, amount, fees where applicable, status, references, and provider references.
- View related user and ledger records.
- Review pending transactions.
- Review failed transactions and available failure reasons.
- Support controlled refund/reversal workflows where applicable.

8. VTU / Airtime / Data Services

- View VTU transaction activity.
- View airtime purchases.
- View data purchases.
- View supported service categories.
- Track successful, pending, and failed VTU transactions.
- View provider transaction/reference IDs.
- View provider response information where appropriate.
- Monitor VTU spending by day/month.
- Identify failed or delayed provider transactions.
- Keep provider API credentials and secrets on the backend only.

9. VTU Provider Balance

- Display the actual wallet/account balance held with the VTU provider when the provider API supports balance lookup.
- Provide a refresh mechanism to retrieve the latest provider balance.
- Display today's provider spending.
- Display monthly provider spending.
- Display provider transaction counts by status.
- Configure a low-balance threshold.
- Show an alert when provider balance falls below the configured threshold.
- Show provider/service status when the provider API supports it.
- Keep provider credentials securely on the backend.

10. Analytics & Reports

- Transaction volume over time.
- Transaction count over time.
- Total transaction value over time.
- Deposit volume over time.
- VTU sales volume and value.
- Successful versus failed versus pending transaction rates.

- Top users by transaction count.
- Top users by transaction value.
- Most-used VTU/service types.
- New user registrations over time.
- Active users over time.
- Application fees/revenue where applicable.
- Custom date-range filtering.
- Export reports where appropriate.

11. Fraud / Suspicious Activity Detection

- Flag multiple transactions within a very short period.
- Flag unusually frequent deposits.
- Flag unusually large transactions.
- Flag rapid deposits followed by unusual activity.
- Flag repeated failed transactions or unusual transaction patterns.
- Allow configurable risk thresholds.
- Display why an activity was flagged.
- Allow an administrator to open a review case.
- Support statuses such as New, Under Review, Resolved, False Positive, and Escalated.
- Allow investigation notes.
- Link risk cases to affected users and transaction IDs.
- Keep a permanent investigation history.
- Risk alerts should be treated as signals for review, not automatic proof of fraud.

12. Customer Support / Issue Handling

- View user-reported financial or service issues.
- Attach support notes to users and transactions.
- Investigate transactions using references and ledger history.
- Document how an issue was resolved.
- Link support cases to suspicious-activity cases when relevant.
- Record which administrator handled the issue and what action was taken.

13. Notifications & Alerts

- Low VTU provider balance alerts.
- High failed-transaction alerts.
- Large-value transaction alerts.
- Suspicious-activity alerts.
- Pending transaction alerts.
- Provider/service outage alerts where supported.
- Unread/read status for admin notifications.

14. Audit Logs

- Record every sensitive administrative action.
- Record administrator identity.
- Record action type.
- Record affected user, transaction, or resource.
- Record timestamp.
- Record relevant before/after values.
- Record reason or note where required.
- Record IP/device/session information where appropriate.
- Prevent ordinary administrators from deleting audit records.
- Provide search and filtering for audit events.

15. Admin Users & Roles

- Create administrator accounts where authorized.
- Activate/deactivate administrator accounts.
- Assign roles.
- Assign individual permissions where supported.
- Restrict sensitive actions such as balance adjustments to authorized roles.
- View administrator activity through audit logs.

16. System Settings

- Configure transaction limits.
- Configure minimum and maximum deposit amounts.
- Configure VTU purchase limits where applicable.
- Configure low provider-balance threshold.
- Configure fraud/risk thresholds.
- Manage admin roles and permissions.
- Configure application fees/charges where supported.
- Configure service availability or maintenance controls where appropriate.

17. Security Requirements

- All privileged operations must be authorized by the backend.
- Never trust frontend-only permission checks.
- Protect admin APIs with authentication, authorization, validation, and rate limiting.
- Never expose VTU/payment/database secrets in the frontend.
- Use idempotency for financial operations to prevent duplicate processing.
- Use atomic database transactions for balance and ledger changes.
- Do not delete financial history.
- Log sensitive administrative actions.

- Protect against broken access control, injection, and unauthorized resource access.
- Require confirmation or re-authentication for especially sensitive actions where appropriate.

18. Recommended Admin Dashboard Navigation

#	Section
1	Dashboard / Overview
2	Users
3	Deposits
4	Transactions
5	VTU / Services
6	Provider Balance
7	Analytics & Reports
8	Fraud / Risk Alerts
9	Support / Cases
10	Notifications
11	Audit Logs
12	Admin Users & Roles
13	Settings

19. Financial Architecture Rule

The ledger/database is the financial source of truth. The admin dashboard is a control and monitoring layer. The frontend must never directly modify balances. All balance changes must pass through secured backend APIs, validation, atomic database operations, ledger entries, and audit logging.

20. Scope Exclusions for This V2

- No user withdrawal functionality.
- No user-to-user transfer functionality.
- No admin withdrawal-management module is required.
- No admin transfer-management module is required.
- If these features are added in a future version, they should be introduced as separate requirements rather than assumed in the current dashboard.